- Dwarf willows, birches, mosses, grasses, sedges are the flora here.
- Reindeer, arctic hare, musk ox, lemmings are important Tundra herbivores. Some important carnivores are the arctic fox, arctic wolf, bobcat and snowy owl. Polar bears live along coastal areas.
- Because of the severe winters, many of the animals are migratory. For example, the many shore birds and waterfowl such as ducks and geese, nest in the Tundra during the summer and migrate south for the winter.

TAIGA BIOME

- The Taiga is 1300-1450 km wide zone south of the Tundra.
- This area has long and cold winters.
- Summer temperature ranges from 10^0 C to 21^0 C.
- Precipitation ranges about 380-1000 mm annually.
- The Taiga is a forest of coniferous trees such as spruce, fir and pine. This is a major source for the logging industry.
- Important migratory herbivores include moose, elk, deer and reindeer. Moose and reindeer migrate to the Taiga for winter and to the Tundra for summers. The common smaller mammals are herbivorous squirrels, snowshoe hare and predatory pine martens. Important predators include the timber wolf, grizzly bear, black bear, bobcat and wolverines. (**Fig. 11.6**)



Fig. 11.6 Taiga biome

GRASSLAND BIOME

- Grasslands occur in temperate and in the tropical regions.
- They have hot summers, cold winters, and irregular rainfall.
- Often they are characterized by high winds.
- The low irregular rainfall is the factor which makes the difference between a temperate deciduous forest and a temperate grassland.
- Herbivores like antelope, bison, wild horse, jack rabbit, ground squirrel and prairie dogs are abundant.
- Predators include coyotes, foxes, hawks and snakes.
- In India, fauna of grasslands includes Elephant, Gaur, Rhino, Antelope.
- Flora of grasslands include purple needle grass, wild oats, foxtail, ryegrass and buffalo grass (**Fig. 11.7).**



Fig. 11.7 Grassland biome

Alpine biome

- The alpine zone (zone between timber line and snow zone) includes in the descending order, a sub-snow zone immediately below the snow zone, a meadow zone in the centre and a shrub zone which gradually merges into the timber zone.
- The snow zone of Himalayas lies over 5100m above mean sea level and alpine zone exists at a height of 3600m. From an ecological view point, the zone above the limits of tree growth (timber line) exhibits extreme environmental conditions which greatly influence the biota of this region.

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